

Bank Customer Churn Analysis

1. Project Overview

This project analyzes customer churn patterns based on 10,000 customers' demographic, credit etc. It consists of data regarding customer demographics like geography, age, gender, along with data like how many years the customer has been with the bank, whether they are an active member, if they have a credit card etc.

2. Dataset Summary

i) **Rows:** 10000

ii) **Columns:** 18

iii) **Missing Values:** There are no missing values within the dataset.

iv) **Key Features:** This dataset contains information regarding the following :-

a) **Customer Demographics** – Age, Gender, Geography

b) **Banking Related** – Has Cr Card, Is Active Member, Card Type, Tenure, Balance, Num of Products

c) **Financial Profile** – Credit Score, Estimated Salary, Balance

d) **Service and Loyalty** – Complain, Satisfaction Score, Point Earned

e) **Target Variable** – Exited

3. Exploratory Data Analysis using Python

We start with some data cleaning and transformation using Python :-

i) **Data Loading:** Imported the data into Python using pandas.

ii) **Initial Exploration:** Used .info() and .describe() to get information about the structure and the statistics of the data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row Number            10000 non-null  int64
1   Customer Id           10000 non-null  int64
2   Surname               10000 non-null  object
3   Credit Score          10000 non-null  int64
4   Geography             10000 non-null  object
5   Gender                10000 non-null  object
6   Age                  10000 non-null  int64
7   Tenure                10000 non-null  int64
8   Balance               10000 non-null  float64
9   Num Of Products       10000 non-null  int64
10  Has Cr Card           10000 non-null  int64
11  Is Active Member      10000 non-null  int64
12  Estimated Salary      10000 non-null  float64
13  Exited                10000 non-null  int64
14  Complain              10000 non-null  int64
15  Satisfaction Score    10000 non-null  int64
16  Card Type             10000 non-null  object
17  Point Earned          10000 non-null  int64
dtypes: float64(2), int64(12), object(4)
memory usage: 1.4+ MB
```

[5]:

	Row Number	Customer Id	Surname	Credit Score	Geography	Gender	Age	Tenure	Balance	Num Of Products	Has Cr Card	Is Active Member	Estimated Salary
count	10000.00000	1.000000e+04	10000	10000.000000	10000	10000	10000.000000	10000.000000	10000.000000	10000.000000	10000.00000	10000.000000	10000.000000
unique	NaN	NaN	2932	NaN	3	2	NaN	NaN	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	Smith	NaN	France	Male	NaN	NaN	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	32	NaN	5014	5457	NaN	NaN	NaN	NaN	NaN	NaN	NaN
mean	5000.50000	1.569094e+07	NaN	650.528800	NaN	NaN	38.921800	5.012800	76485.889288	1.530200	0.70550	0.515100	100090.239881
std	2886.89568	7.193619e+04	NaN	96.653299	NaN	NaN	10.487806	2.892174	62397.405202	0.581654	0.45584	0.499797	57510.492818
min	1.00000	1.556570e+07	NaN	350.000000	NaN	NaN	18.000000	0.000000	0.000000	1.000000	0.00000	0.000000	11.580000
25%	2500.75000	1.562853e+07	NaN	584.000000	NaN	NaN	32.000000	3.000000	0.000000	1.000000	0.00000	0.000000	51002.110000
50%	5000.50000	1.569074e+07	NaN	652.000000	NaN	NaN	37.000000	5.000000	97198.540000	1.000000	1.00000	1.000000	100193.915000
75%	7500.25000	1.575323e+07	NaN	718.000000	NaN	NaN	44.000000	7.000000	127644.240000	2.000000	1.00000	1.000000	149388.247500
max	10000.00000	1.581569e+07	NaN	850.000000	NaN	NaN	92.000000	10.000000	250898.090000	4.000000	1.00000	1.000000	199992.480000

Exited	Complain	Satisfaction Score	Card Type	Point Earned
10000.000000	10000.000000	10000.000000	10000	10000.000000
NaN	NaN	NaN	4	NaN
NaN	NaN	NaN	DIAMOND	NaN
NaN	NaN	NaN	2507	NaN
0.203800	0.204400	3.013800	NaN	606.515100
0.402842	0.403283	1.405919	NaN	225.924839
0.000000	0.000000	1.000000	NaN	119.000000
0.000000	0.000000	2.000000	NaN	410.000000
0.000000	0.000000	3.000000	NaN	605.000000
0.000000	0.000000	4.000000	NaN	801.000000
1.000000	1.000000	5.000000	NaN	1000.000000

iii) Checked for Null Values: Checked if there were any null values, there weren't any, so no imputation needed.

iv) Standardized Column Names: Standardized the names of the columns to snake case, for better documentation and readability.

[8]:

```
Index(['row_number', 'customer_id', 'surname', 'credit_score', 'geography',
      'gender', 'age', 'tenure', 'balance', 'num_of_products', 'has_cr_card',
      'is_active_member', 'estimated_salary', 'exited', 'complain',
      'satisfaction_score', 'card_type', 'point_earned'],
      dtype='object')
```

v) Data Quality Check: Dropped unnecessary columns like Row Number and Surname.

vi) Feature Engineering: Created 2 new columns :-

a) **credit_score_range** – Created from the **credit_score** column.

b) **age_group** – Grouped together customers based on their ages.

vii) Checked for Duplicate values – There weren't any duplicates within the data.

viii) Database Connection – Loaded the transformed data into PostgreSQL using psycopg2 and sqlalchemy, for further analysis.

4. Data Analysis using SQL

Performed some analysis using PostgreSQL to answer some business questions :-

i) No of Active Customers who Churn

	exited bigint	no_of_active_churners bigint
1	0	4416
2	1	735

ii) No of Churners and Non-Churners by Gender

	gender text	exited bigint	no_of_customers bigint
1	Female	0	3404
2	Female	1	1139
3	Male	0	4558
4	Male	1	899

iii) No of Churners and Non-Churners by Geography

	geography text	exited bigint	no_of_customers bigint
1	France	0	4203
2	France	1	811
3	Germany	0	1695
4	Germany	1	814
5	Spain	0	2064
6	Spain	1	413

iv) Customer Churn by Age Group

	age_group text	no_of_churners bigint
1	Adult	249
2	Middle-Aged	532
3	Senior	1044
4	Young Adult	213

v) No of Churners and Non-Churners by Tenure

	tenure bigint	total_customers bigint	no_of_churners bigint	no_of_nonchurners bigint
1	0	413	95	318
2	1	1035	232	803
3	2	1048	201	847
4	3	1009	213	796
5	4	989	203	786
6	5	1012	209	803
7	6	967	196	771
8	7	1028	177	851
9	8	1025	197	828
10	9	984	214	770
11	10	490	101	389

vi) Customer Churn based on the number of products they use from the Bank

	num_of_products bigint	total_customers bigint	no_of_churners bigint	churn_rate numeric
1	1	5084	1409	27.71
2	2	4590	349	7.60
3	3	266	220	82.71
4	4	60	60	100.00

vii) No of Customers and Churners by Complaint Status

	complain bigint	no_of_customers bigint	no_of_churners bigint
1	0	7956	4
2	1	2044	2034

viii) Customer Churn based on whether or not they own a Credit Card

	credit_card_owner text	no_of_churners bigint
1	No	613
2	Yes	1425

ix) Customer Churn based on Balance (highest, lowest and average balance)

	balance_segment text	no_of_churners bigint	avg_balance numeric	highest_balance double precision	min_balance double precision
1	Medium	987	123693.73	149995.72	100013.51
2	Zero	500	0.00	0	0
3	Low	327	81763.91	99986.98	3768.69
4	High	224	166767.67	250898.09	150005.38

x) Top 3 Age Groups by Churn Rate in each Country

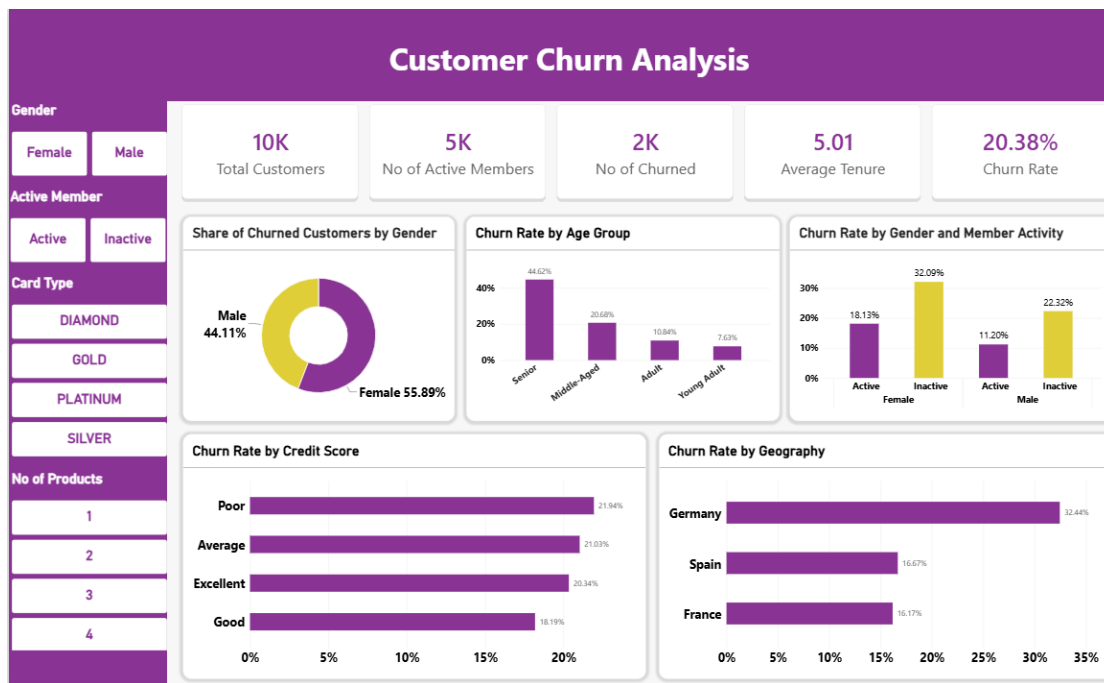
	geography text	age_group text	churn_rate numeric
1	France	Senior	38.70
2	France	Middle-Aged	17.02
3	France	Adult	8.33
4	Germany	Senior	61.63
5	Germany	Middle-Aged	32.15
6	Germany	Adult	19.40
7	Spain	Senior	35.31
8	Spain	Middle-Aged	16.36
9	Spain	Young Adult	8.47

xi) Churn Rate by Balance Quartile within Each Country

	geography text	balance_quartile integer	no_of_churners bigint	churn_rate numeric
1	France	1	177	14.11
2	France	2	194	15.47
3	France	3	196	15.64
4	France	4	244	19.47
5	Germany	1	140	22.29
6	Germany	2	277	44.18
7	Germany	3	238	37.96
8	Germany	4	159	25.36
9	Spain	1	81	13.06
10	Spain	2	97	15.67
11	Spain	3	116	18.74
12	Spain	4	119	19.22

5. Dashboard in Power BI

Finally, we built an interactive dashboard to visualize our data, to get better insights.



6. Business Recommendations

i) Gender Targeted Efforts – Female customers make up 55.89% of all churners vs 44.11% for male customers, investigate the reason for it and apply some targeted retention schemes based on customer behavior and needs.

ii) Product Depth Strategy – Customers with 2 products tend to have the lowest churn rate, 7.6%, while those who purchase 3 or 4 products have the highest churn rate, 82.71% and 100%. Though the last 2 categories have lesser amount of customers, it is still worth investigating the cause behind this (for example – price, fees, product fit etc.), before promoting any selling further than 2 products.

iii) Region Based Efforts – Germany's churn rate is 32.44% as compared to Spain and France's 16.67% and 16.17%, look into the reasons for it and rectify it using some targeted retention strategies, and track the churn rate quarterly.